ANTHONY ROLLAND

Backend Software Engineer

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Summary

Back-end Software Engineer with hands-on production experience building Spring Boot APIs and microservices on AWS, currently apprenticing at SNCF Connect & Tech. Actively building a RAG system with Spring AI and pgvector. Seeking an international 12–14 week Software Engineering internship for Spring/Summer 2027. Graduating in August 2028.

Work Experience

SNCF Connect&Tech | Apprentice Software Engineer | Java, Spring Boot, AWS, PostgreSQL

Sep 2025 - Aug 1

- Develop and maintain Spring Boot REST APIs managing reference data for France's 3,000+ train stations and 300,000+ equipment records, backed by PostgreSQL and deployed on AWS.
- Reduced CSV export time by 45% in a station-tracks management interface (Java, Set-based deduplication)
- Maintained >80% test coverage using JUnit 5 and Mockito; enforced SonarQube quality gates to sustain Grade A code quality across all critical services, with active participation in peer code reviews.

Meelo - Fintech Startup | Back-End Developer Internship | Java, Spring Boot, PostgreSQL

Apr 2024 - Jun 2024

- Designed and shipped a standalone notification microservice to production integrating with existing Spring Boot services.
- Optimized PostgreSQL database performance by refactoring queries and implementing indexing strategies, reducing query latency by ~10%.

Projects

Social Network Platform | Spring Boot, MongoDB, Next.js, React Native, Hugging Face

Oct 2024 - Jan 2025

- Designed and owned the back-end API architecture using Spring Boot and MongoDB, implementing JWT authentication and deploying the full stack to a school environment via Terraform.
- Integrated AI models from Hugging Face for dynamic content generation (avatars/posts) and engineered a real-time messaging system.

RAG API | Spring Boot, Spring AI, Ollama, pgvector, PostgreSQL, AWS S3

Feb 2026 - Present

- Engineered a semantic document Q&A API using Spring Boot, Spring AI, pgvector, and Ollama.
- Built an async ingestion pipeline (AWS S3, PostgreSQL, nomic-embed-text) processing up to 150-page documents with Flyway-managed schema migrations.
- Containerized the full stack with Docker Compose using multi-stage builds and HNSW vector indexing.

Technical Skills

Languages: Java, Python, TypeScript, SQL, Bash | Databases: PostgreSQL, MongoDB

Frameworks: Spring Boot, Spring Data JPA, Spring AI, Hibernate

DevOps & Cloud: Docker, AWS, Linux, CI/CD, Git, Gradle, Terraform

Concepts: Microservices, System Design, REST APIs, AI Integration, RAG & LLMs

Education

Bachelor of Science in Computer Science — IUT de Lens | Sep 2022 - Jul 2025

Master of Science in Computer Science — INSA Hauts-de-France | Sep 2025 - Aug 2028

Languages: French (Native), English C1 - TOEIC 970/990, German (Basic)